

MTD2A_binary_output

MTD2A: Model Train Detection And Action – Arduino library <https://github.com/MTD2A/MTD2A>
Jørgen Bo Madsen / V1.2.1 / 05-07-2026

Model Train Detection And Action is a collection of user friendly advanced and functional C++ building blocks for time controlled handling of input and output, using a state machine system for parallel processing. The library is intended for Arduino enthusiasts without much programming experience who are interested in electronics control and automation, as well as model trains.

Common to all building blocks:

- Build on a state machine system for parallel processing and synchronous time management
- They support a wide range of input sensors and output devices
- Are simple to use to build complex solutions with few commands
- They operate non-blocking, process-oriented and state-driven
- Offers extensive control and troubleshooting information
- Thoroughly documented with many examples

Table of contents

MTD2A_binary_output	1
Feature Description	1
Process phases	3
Initialization	4
Activation.....	4
Stop and reset (restart) timer	5
outputValue	6
PWM curves.....	7
Scaling	13
outputValue and PWM	14
Examples of configuration	16
Set get functions	17
print_conf();.....	18

Feature Description

MTD2A_binary_output process consists of 4 functions:

1. `MTD2A_binary_output object_name ("object_name" ,
outputTimeMS, beginTimeMS, endTimeMS, { BINARY | PWM }, beginValue, endValue, loopActivate);`
2. `object_name.initialize ( pinNumber, pinWriteMode, startPinValue );`
Executed once in `void setup ();` and after `Serial.begin (9600);`
3. `object_name.activate();` Activate once and only work again when the process is completed **(COMPLETE)**
4. `MTD2A_loop_execute ();` Called as the last instruction in `void loop ();`

All functions use default values and can therefore be called with none and up to eight parameters.

However, parameters must be specified in ascending order. See example below:

```
MTD2A_binary_output object_name;  
MTD2A_binary_output object_name  
("object_name");  
("object_name", outputTimeMS);  
("object_name", outputTimeMS, beginTimeMS);  
("object_name", outputTimeMS, beginTimeMS, endTimeMS);  
("object_name", outputTimeMS, beginTimeMS, endTimeMS, outputMode);  
("object_name", outputTimeMS, beginTimeMS, endTimeMS, outputMode, beginValue);  
("object_name", outputTimeMS, beginTimeMS, endTimeMS, outputMode, beginValue, endValue);  
("object_name", outputTimeMS, beginTimeMS, endTimeMS, outputMode, beginValue, endValue, loopActivate);
```

Defaults:

```
("output_name", 0, 0, 0, BINARY, HIGH, LOW, DISABLE);
```

Example

```
// Two blinking LEDs. One with symmetric interval and another with asymmetric interval.  
  
#include <MTD2A.h>  
using namespace MTD2A_const;  
  
MTD2A_binary_output red_LED ("Red LED", 400, 400); // 0.4 sec light, 0.4 sec no light  
MTD2A_binary_output green_LED ("Green LED", 300, 700, 0, P_W_M, 96); // 0.3 / 0.7 sec PWM dimmed  
  
void setup() {  
    Serial.begin(9600);  
    while (!Serial) { delay(10); } // ESP32 Serial Monitor ready delay  
  
    byte RED_LED_PIN = 9; // Arduino board pin number  
    byte GREEN_LED_PIN = 10; // Arduino board pin number  
    red_LED.initialize (RED_LED_PIN);  
    green_LED.initialize (GREEN_LED_PIN);  
    Serial.println("Two LED flashes");  
}  
  
void loop() {  
    if (red_LED.get_processState() == COMPLETE) {  
        red_LED.activate();  
    }  
    if (green_LED.get_processState() == COMPLETE) {  
        green_LED.activate();  
    }  
  
    MTD2A_loop_execute();  
}
```

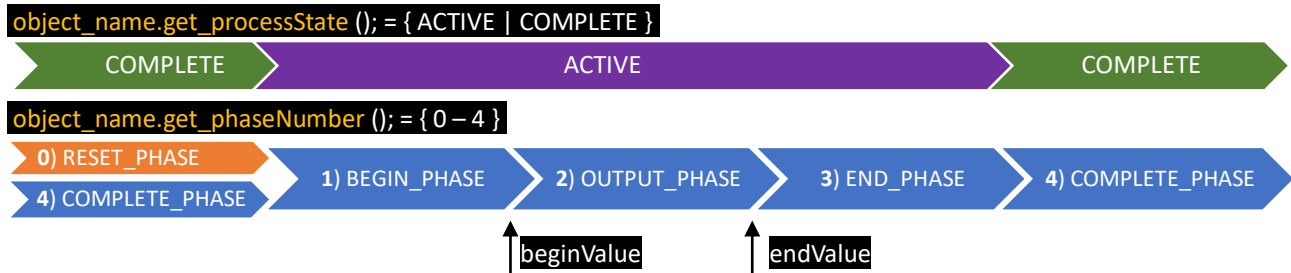
More examples and Youtube video:

<https://github.com/MTD2A/MTD2A/tree/main/examples>

DEMO video: <https://youtu.be/eyGRazX9Bko>

Process phases

Depending on the current configuration, the process is carried out in between 1 and 5 stages.



0. The initial phase when the program starts and when the function `reset();` is called or 4) **COMPLETE_PHASE**.
1. Start delay. If set to 0 or not defined, the stage will be skipped.
2. Output time period. Starts with **beginValue** and ends with **endValue**. If set to 0 or not defined, the stage will be skipped, and a warning is written.
3. End of delay. If set to 0 or not defined, the stage will be skipped.
4. The process is completed **COMPLETE** and ready for reactivation `object_name.activate();`

Global number constants: **RESET_PHASE**, **BEGIN_PHASE**, **OUTPUT_PHASE**, **END_PHASE** & **COMPLETE_PHASE**

The immediate phase shift can be identified by function: `object_name.get_phaseChange(); = { true | false }`

Process status

When transitioning to **BEGIN_PHASE**, **OUTPUT_PHASE**, or **END_PHASE**, ProcessState switches to **ACTIVE**.

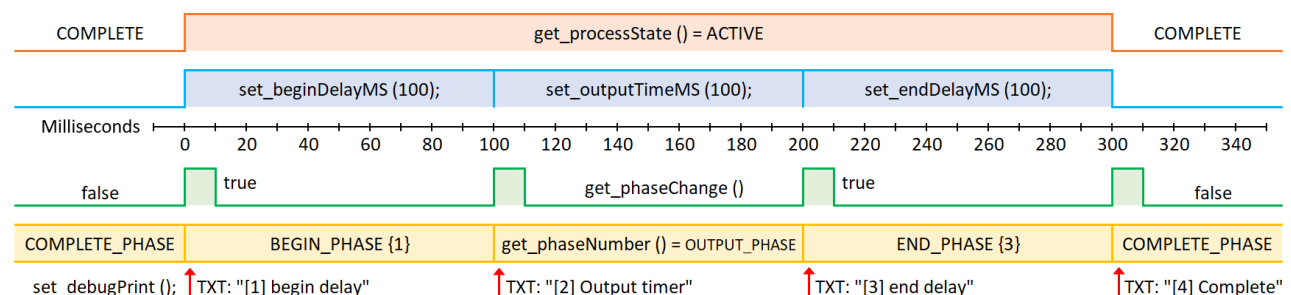
When transitioning to **COMPLETE_PHASE** or **RESET_PHASE**, the processState switches to **COMPLETE**.

The **immediate** phase shift can be identified by function: `object_name.get_phaseChange(); = { true | false }`

Status information can be printed with the function: `object_name.set_debugPrint(); = { ENABLE | DISABLE }`

Example of different configuration methods

```
MTD2A_binary_output relay ("Relay", 100, 100, 100, BINARY, HIGH, LOW);
// is equivalent to
MTD2A_binary_output relay ("Relay", 100, 100, 100); // default: BINARY, HIGH, LOW
// is equivalent to
MTD2A_binary_output relay ("Relay"); // default: 0, 0, 0, BINARY, HIGH, LOW
relay.set_outputTimeMS (100);
relay.set_beginDelayMS (100);
relay.set_endDelayMS (100);
// Show phase change information
relay.set_debugPrint (); // default enable
```



Timing

See the document MTD2A.PDF, section "Cade"

nce" and "Synchronization" as well as "Execution speed".

Initialization

`object_name.initialize ( pinNumber, pinWriteMode, startPinValue );` Is executed only once in `void setup ();` and after `Serial.begin (9600);`. This is important for printing possible errors and warnings, including from previous instantiating of objects.

The output is written to the digital PIN connection number specified in `object_name.initialize ( pinNumber );`. If an unsupported PIN connection is specified, the pin connection will not be initialized, and it will **not** be possible to write to the pin connection subsequently: `pinWriteToggle = DISABLE` and `pinNumber = NO_PIN`. `object_name.initialize ();` is equivalent to `object_name.initialize ( NO_PIN );` `NO_PIN = 255`.

The output on the pin connection can be inverted (reversed) by setting the `pinWriteMode` to `INVERTED`. This means that HIGH and LOW are reversed and the PWM value is calculated to 255 minus PWM value. `object_name.initialize ( pinNumber, {NORMAL | INVERTED} );`

The PIN connection is initialized with the value specified in the `startPinValue`. `startPinValue` is written **instantly** to `pinNumber`. If the parameter is omitted, `LOW` is specified.

`object_name.initialize ( { pinNumber | NO_PIN }, {NORMAL | INVERTED}, {startPinValue | LOW} );`

It is possible to continuously control whether or not to write to the PIN connection with the function:

`object_name.set_pinWriteToggle ( {ENABLE | DISABLE} );`

It is also possible to write directly to the PIN connection with the functions:

`m object_name.set_pinW`

`riteValue ( PinValue ); {HIGH | LOW} / PWM {0-255}`

`object_name.set_pinWriteValue ( PinValue, {BINARY | P_W_M} );` Optional parameter. No default value.

This happens **instantly**, regardless of whether the process is active or not, and independently of `MTD2A_loop_execute ();`

As a starting point, the PIN connection is **undefined**. See [Digital Pins | Arduino Documentation](#)

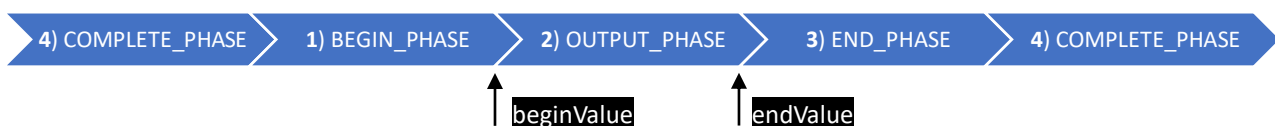
Activation

The process is activated with the function: `object_name.activate ();` This switches the `process State` to **ACTIVE**. As soon as the `processState` switches to **COMPLETE**, the process can be activated again.

The process can be reset at any time with the function: `object_name.reset();` The function resets all control and process variables **instantly**, and before `MTD2A_loop_execute ();`, and prepares for a fresh start. All functionally configured variables and default values are retained. The process phase switches to **RESET_PHASE**.

If the PIN connection is defined correctly, the `startPinValue` is written to the PIN connection **instantly**.

It is possible to write beginning values `beginValue` and end values `endValue` to the PIN connection.



Activate functions use "function overloading". This means that the function can be called with none and up to 4 parameters. However, parameters must be specified in ascending order, starting from the first.

Default values are **not** used. All existing values remain the same unless values are specified as parameters in the function call. See example below:

`object_name.activate ();`

`object_name.activate (beginValue);`

`object_name.activate (beginValue, endValue);`

```
object_name.activate (beginValue, endValue, PWMcurveType);
```

```
object_name.activate (beginValue, endValue, PWMcurveType, outputTimeMS);
```

PWMcurveType specifies which curve to follow in the time period **outputTimeMS**. The curve starts with the value **beginValue** and ends with the value **endValue**.

outputTimeMS indicates the time span between **beginValue** and **endValue** in milliseconds.

Activation can be configured to be repeated indefinitely with function:

object_name.set_loopActivate ({ ENABLE | DISABLE }); Repeats a process started with **activate ()**, enabling it does not start the process. This feature is particularly suitable for flashing LEDs and where a constant pulsing output is required.

When continuous looping is enabled, the process runs through all phases in every cycle, including phase **COMPLETE_PHASE**. The Complete phase lasts one full loop iteration, after which the next cycle starts immediately — the timing of the cycles is not affected. Note that **get_processState ()** returns **ACTIVE** during the whole loop cycle and only returns **COMPLETE** momentarily at each cycle boundary — and permanently after looping is stopped with **object_name.loopActivate (DISABLE)**, which lets the current cycle finish before stopping.

Stop and reset (restart) timer

It is possible to set the new start time for the current timer process, and stop the current timer process prematurely.

RESET_TIMER Sets a new start time which is retrieved from the globally synchronized time **globalSyncTimeMS**.

STOP_TIMER interrupts the time control process

```
object_name.set_outputTimer ( {STOP_TIMER | RESET_TIMER} );
```

```
object_name.set_beginTimer ( {STOP_TIMER | RESET_TIMER} );
```

```
object_name.set_endTimer ( {STOP_TIMER | RESET_TIMER} );
```

The new start time is retrieved from the globally synchronized time and can be read with the function:

```
MTD2A::get_globalSyncTimeMS ();
```

The time period is set by default when the object is instantiated (activated). It is possible to define new time periods for all three time periods:

```
object_name.set_outputTimeMS ( {0 - MAX_TIME_MS} );
```

```
object_name.set_beginDelayMS ( {0 - MAX_TIME_MS} );
```

```
object_name.set_endDelayMS ( {0 - MAX_TIME_MS} );
```

object_name.set_timers combines all three timers into one function. "function overloading" is used. This means that the function can be called with one and up to 3 parameters. However, parameters must be specified in ascending order, starting from the first.

Default values are **not** used . All existing values remain the same unless they are specified as parameters in the function call. See example below:

```
object_name.set_timers (outputTimeMS);
```

```
object_name.set_timers (outputTimeMS, beginDelayMS);
```

```
object_name.set_timers (outputTimeMS, beginDelayMS, endDelayMS);
```

outputValue

Write to `outputValue` and the PIN connection happens in the same millisecond the `MTD2A_loop_execute ();` is started. **But** the value can only be read with `object_name.get_outputValue ();` After `MTD2A_loop_execute ();` is completed, and thus only at the next pass.

There are therefore two logical status flags that can be used to identify when there is data to be read:

1. `object_name.get_outputState ();` Is `true` every time there is new data that can be read.
1. `object_name.get_outputProcess ();` Is `true` when data can be read. That is, throughout the output period.

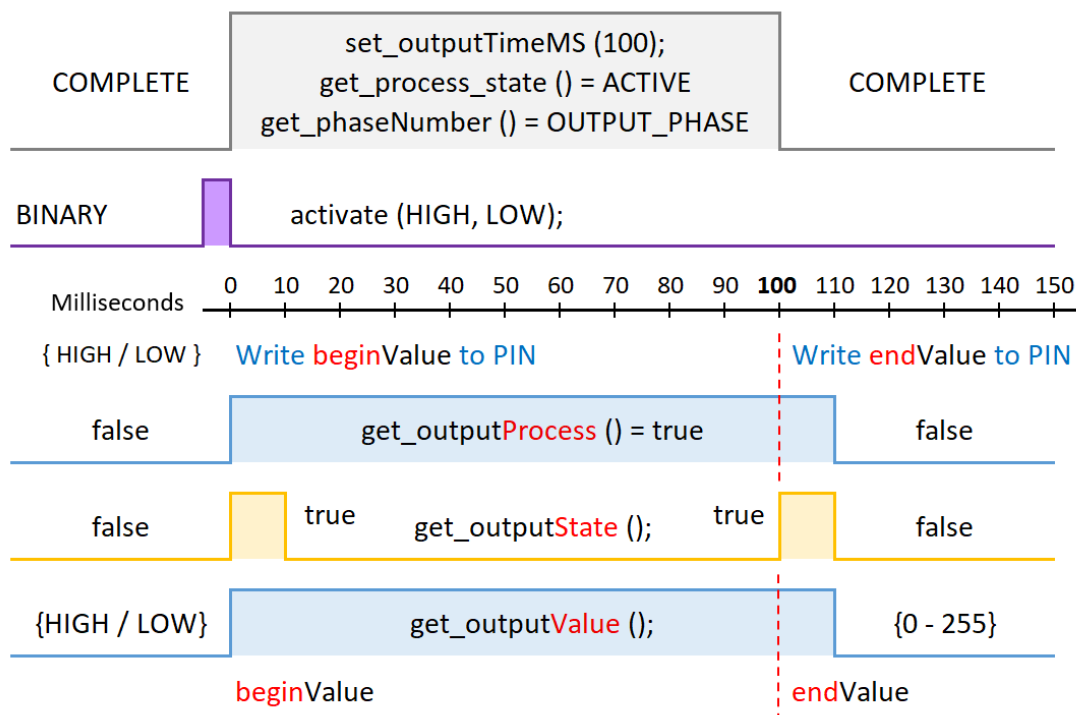
The output value for the PIN connection is specified regardless of whether the PIN connection is defined or not. See explanation of initialization without specifying PIN connection under the section initialization.

The value that can be read from `object_name.get_outputValue ();` is always the last specified value.

NOTE: If `object_name.pinWriteMode` is set to `INVERTED`, this is the inverted value — in !value `BINARY` mode and `255 - value` in `P_W_M` mode — not the value passed to `object_name.activate ();` or `object_name.get_outputValue ();` With the default `object_name.pinWriteMode ( NORMAL );` the two are identical.

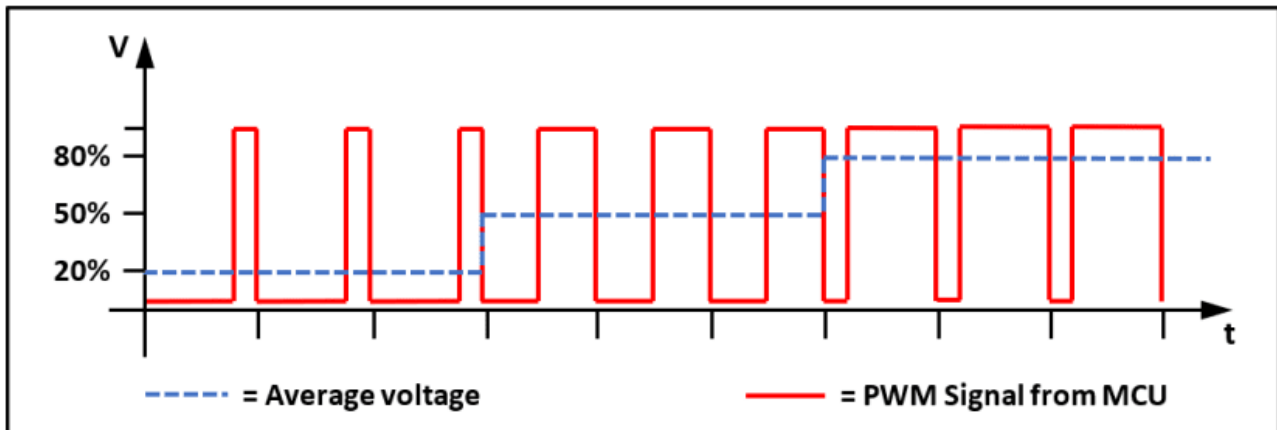
Example:

```
if (relay.get_outputState() == true) {  
    outputValue = relay.get_outputValue();  
    Serial.print("PIN output value: "); Serial.println (outputValue);  
}
```



PWM curves

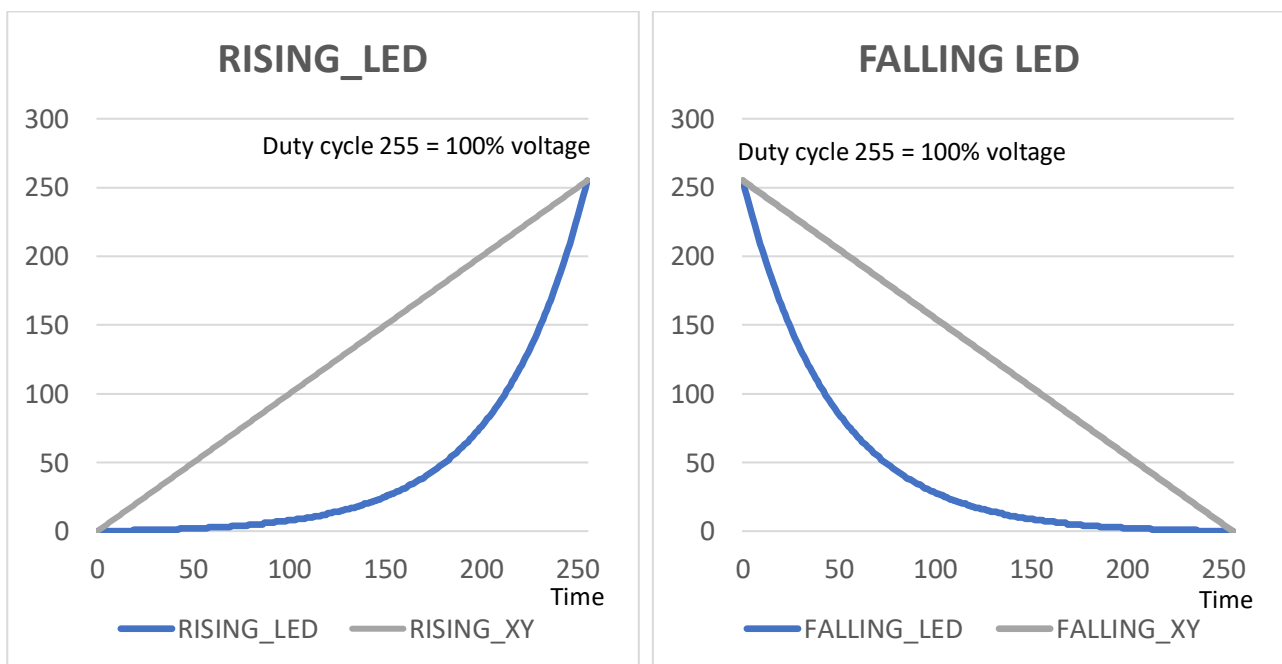
PWM ([Pulse Width Modulation](#)) is a technique used to control the average power delivered to an electrical device by varying the width of a series of pulses. Essentially, it's a digital method of creating an analog effect by rapidly switching a signal on and off



24 different mathematical [PWM](#) curves are designed: 2 linear, 4 exponential, 10 power, 4 Bézier and 4 Sine functions. They are intended to smooth out physical conditions that have a nonlinear relationship between time and function. E.g. acceleration and deceleration DC motors (trains) and servo.

LED fading

PWM is a good way to control the brightness of LEDs. However, the relationship between the duty cycle and the perceived brightness is not linear at all. Humans perceive the brightness change non-linearly. The human eye responds to light in a logarithmic fashion, and has a better sensitivity at low luminance than high luminance.



Arduino example: [MTD2A/examples/math_fade_LED at main · MTD2A/MTD2A](#)

DEMO video: <https://youtu.be/8TV6nOdXBno>

Power functions

Power function is for smoothing acceleration and deceleration.

For example, using power functions is suitable for shuttle service with model trains. The locomotive accelerates smoothly up to speed, runs at a fixed speed for a while, decelerates smoothly down to speed and comes to a complete stop. After a short pause, the process is repeated in the opposite direction.

Inertia and friction

Starting and stopping of train (locomotive and cars) involves both inertia and friction.

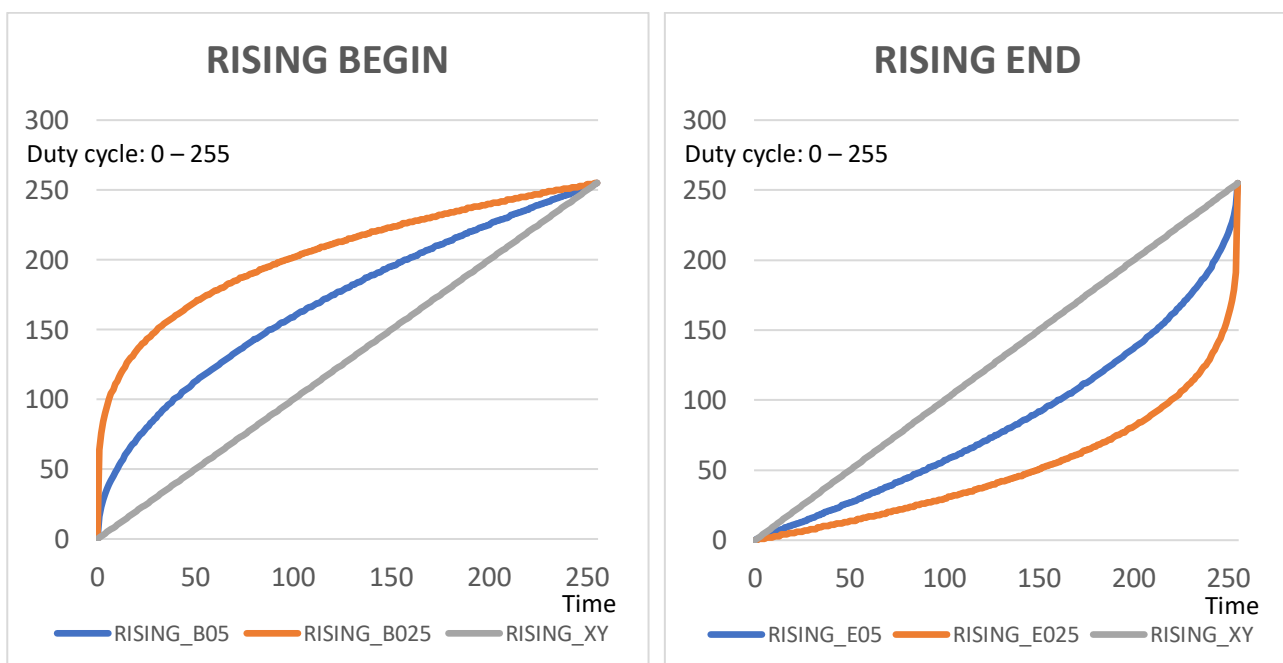
Inertia is the property of an object that causes it to resist changes to its state of motion. To start or stop motion, an unbalanced force is needed to overcome inertia and, in most real-world scenarios, friction.

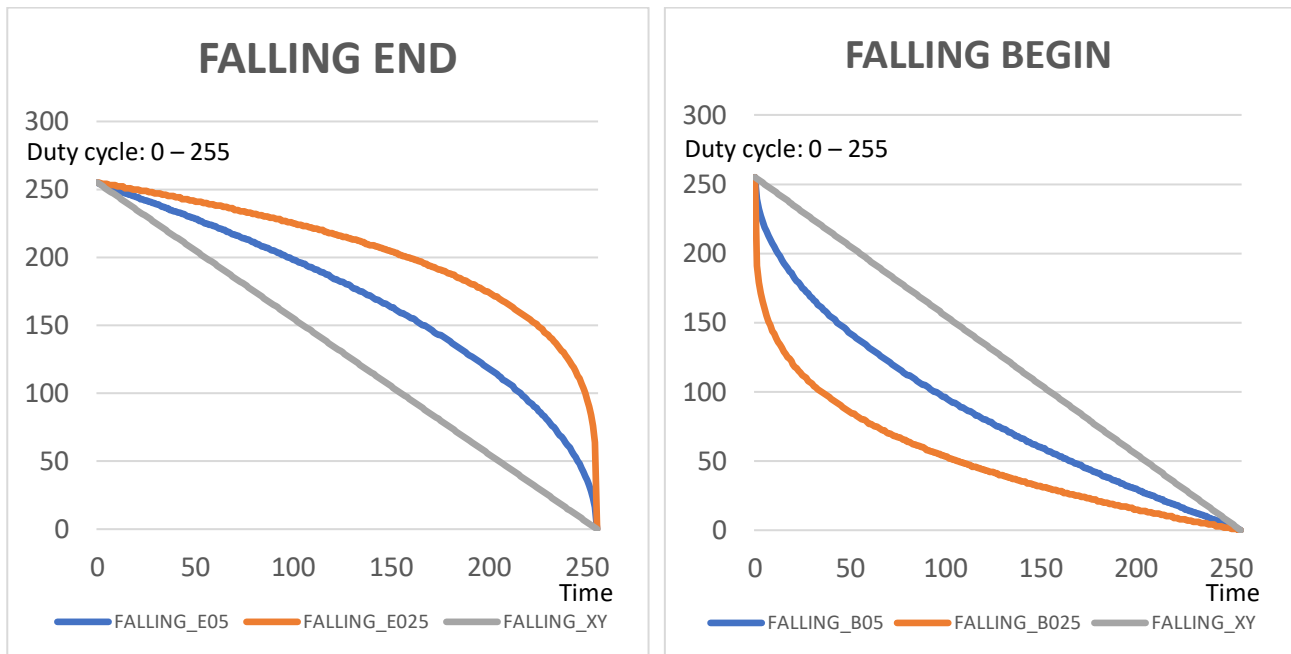
- A car accelerating requires a force to overcome the inertia of the car, which wants to remain stationary.
- A rolling ball eventually stops due to friction, but without friction, it would continue rolling indefinitely.
- Pushing a box: You need to push with enough force to overcome static friction and get the box moving. Once moving, you need to continue applying force to overcome kinetic friction and maintain the speed.

Inertia and friction are fundamental concepts in physics that explain why objects move (or don't move) and how motion can be started and stopped, and the relationship between time and function is **nonlinear**.

Furthermore there can be an effect in DC brushed motors called "armature reaction." Depending on how the motor is manufactured there can be an asymmetry between forward and reverse rotation.

The following 8 power functions will result in more smooth train acceleration and deceleration.

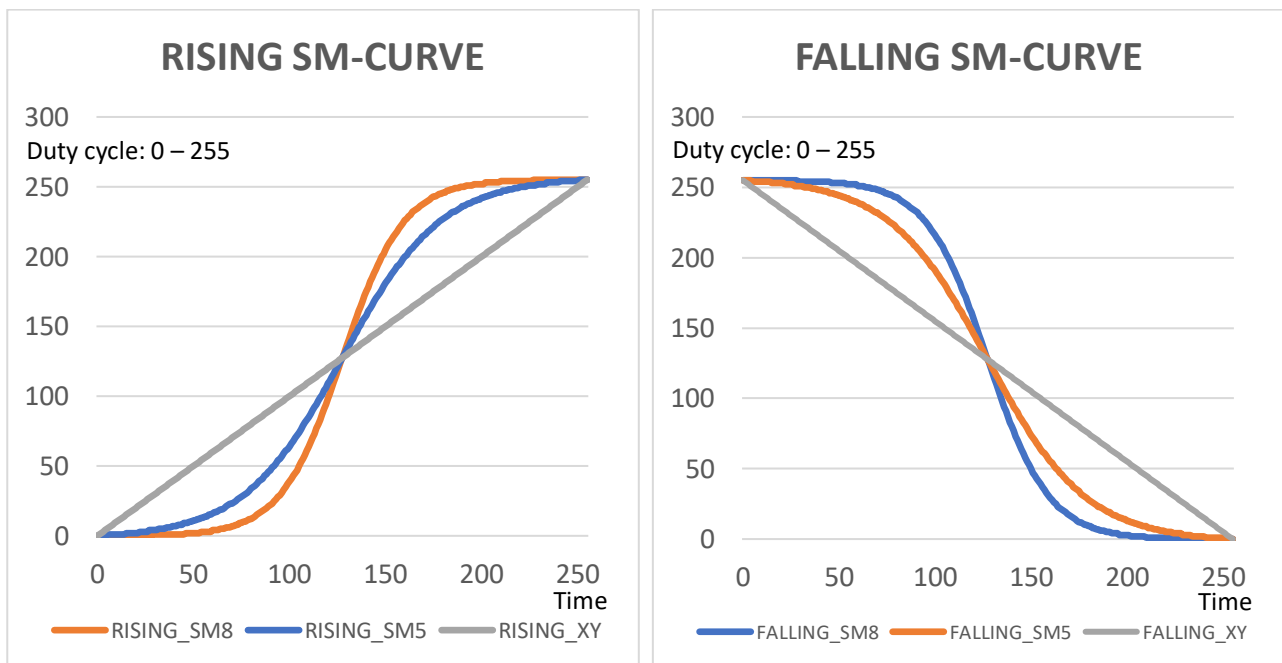




Arduino example: [MTD2A/examples/PWM_power_curves at main · MTD2A/MTD2A](#)
 DEMO video: <https://youtu.be/Fi9D1hrzT9M>

S-curves

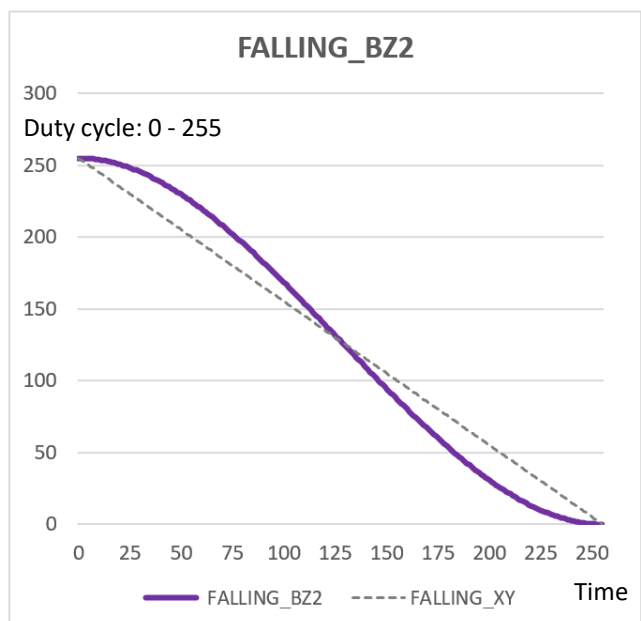
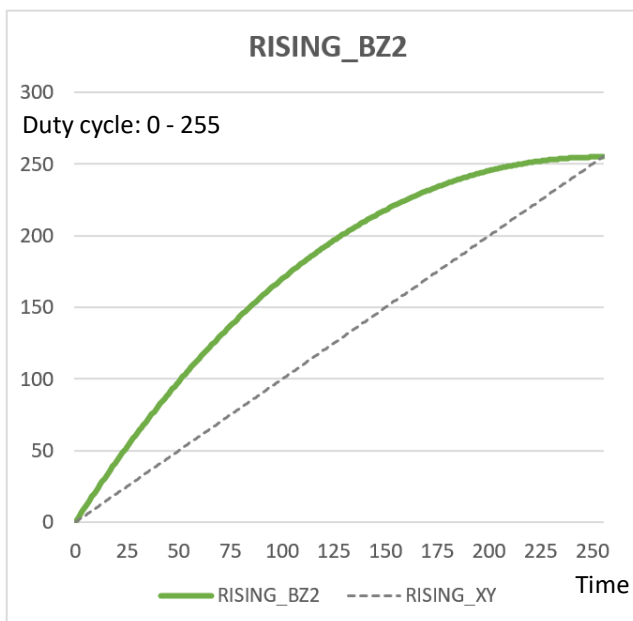
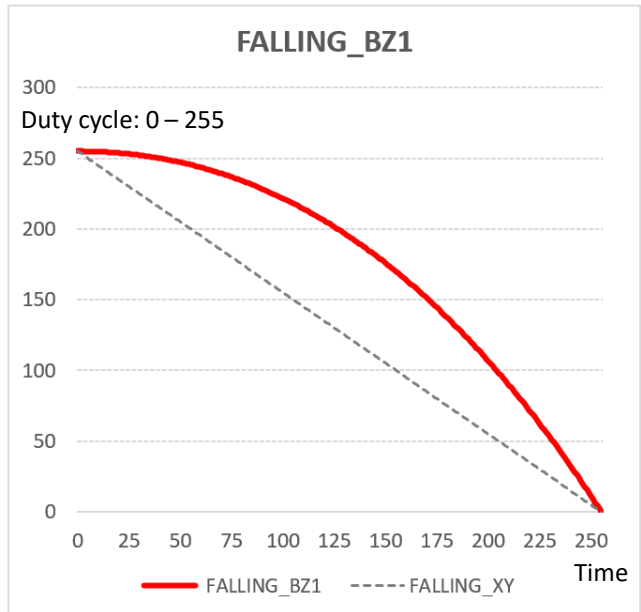
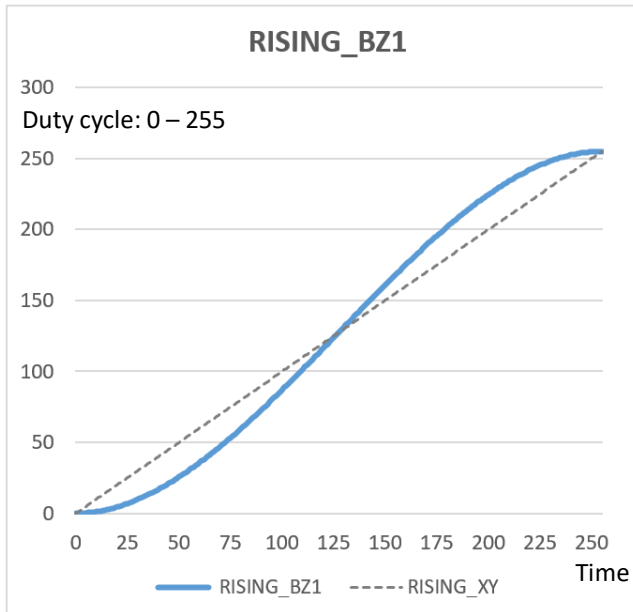
[Sigmoid S-curves](#) can be used where soft acceleration and deceleration is preferable. For example, to control a (heavily loaded) servo motor from point a to b, or a soft start acceleration of a locomotive and a soft end deceleration to continued constant speed.

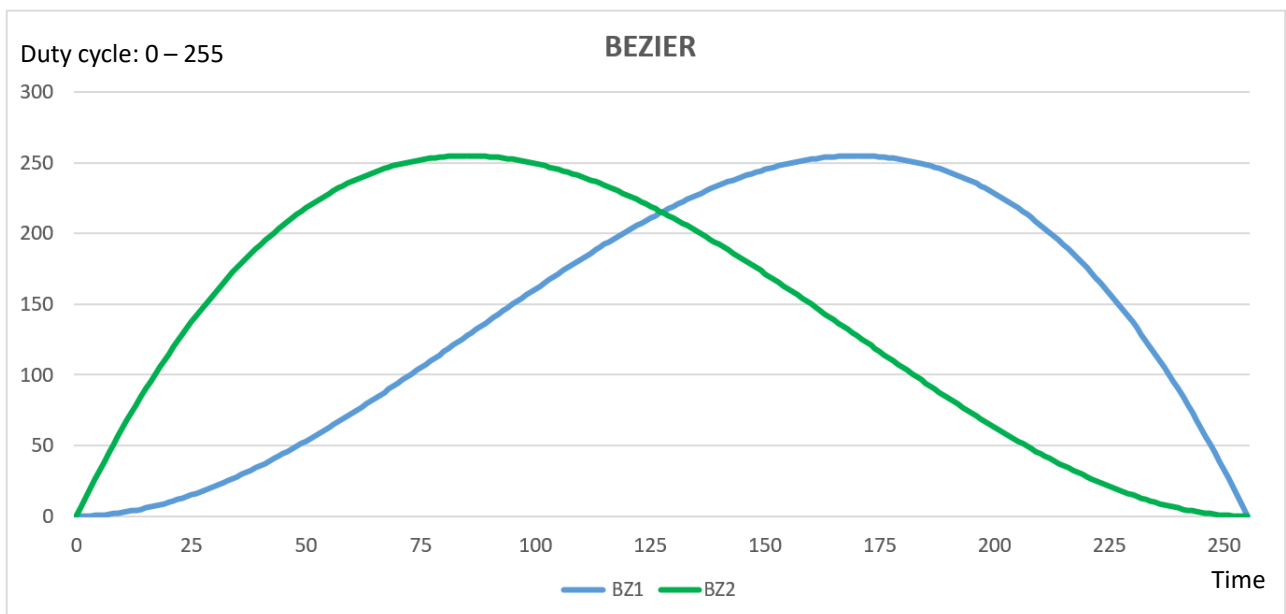


Arduino example: [MTD2A/examples/servo_math_curve at main · MTD2A/MTD2A](#)
 DEMO video: <https://youtu.be/rhQtu0iKFI8>

Bézier curves

[Bézier curves](#) resemble [Sigmoid S-curves](#) and can be used where a smoother acceleration and deceleration is preferred. For example, to control a (heavily loaded) servo motor from point a to b, or a soft start acceleration of a locomotive and a soft end deacceleration to continued constant speed.

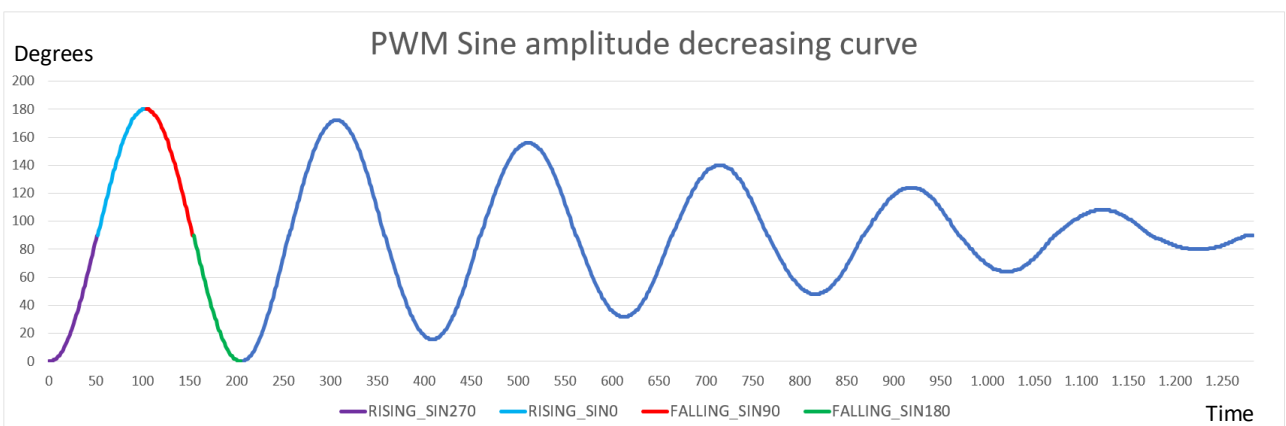
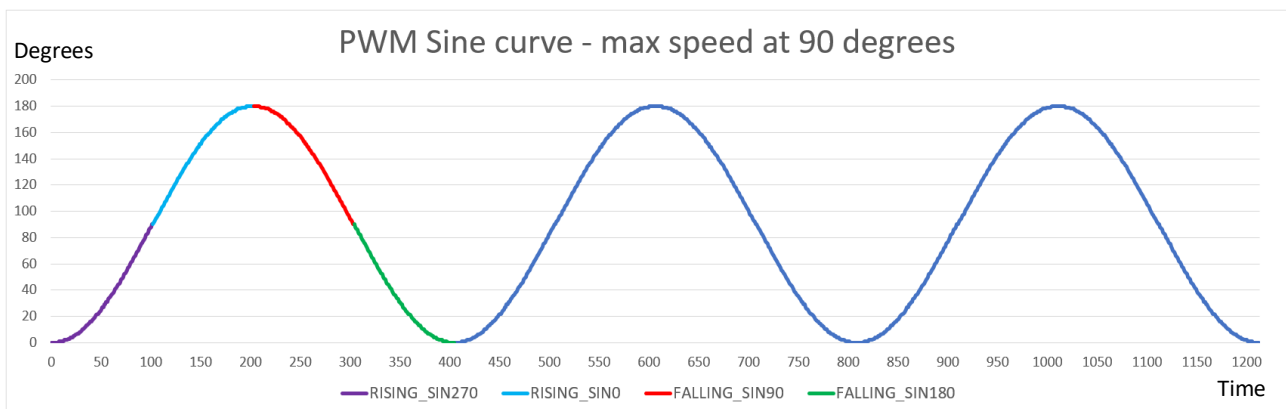


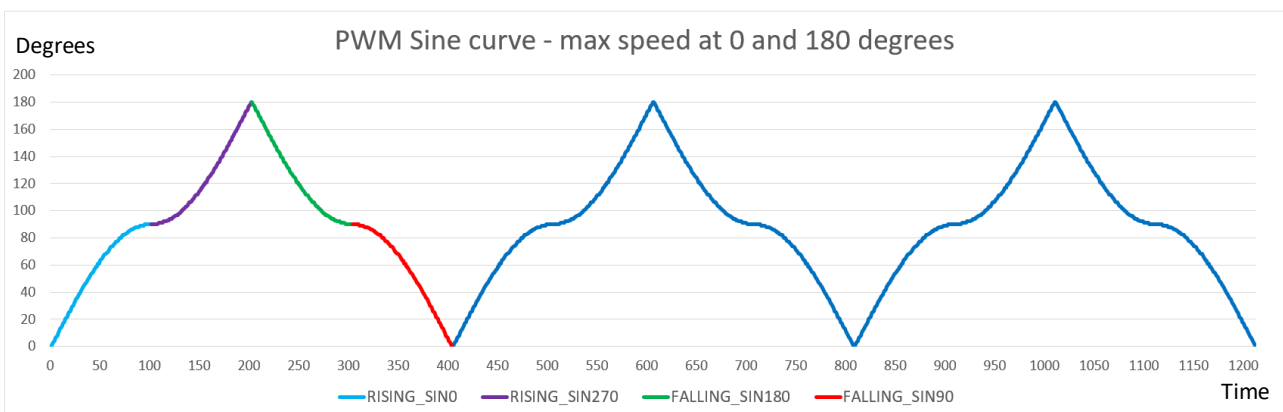
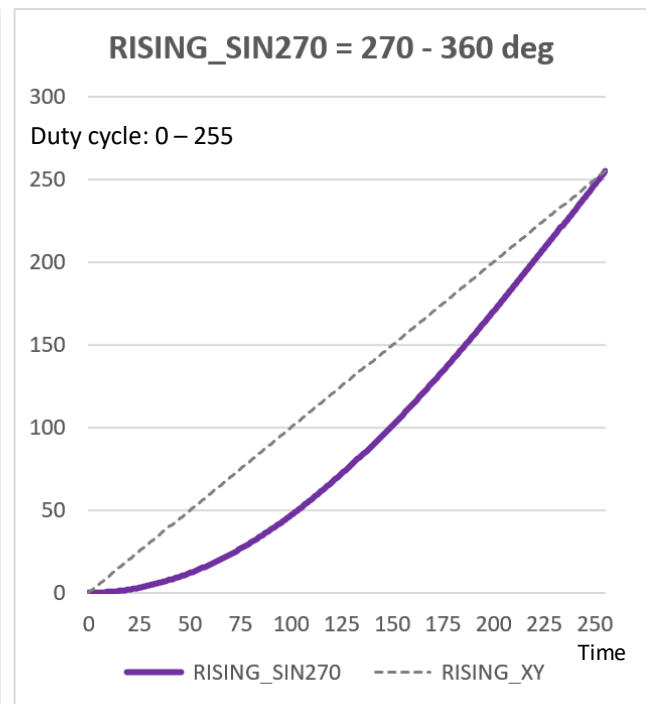
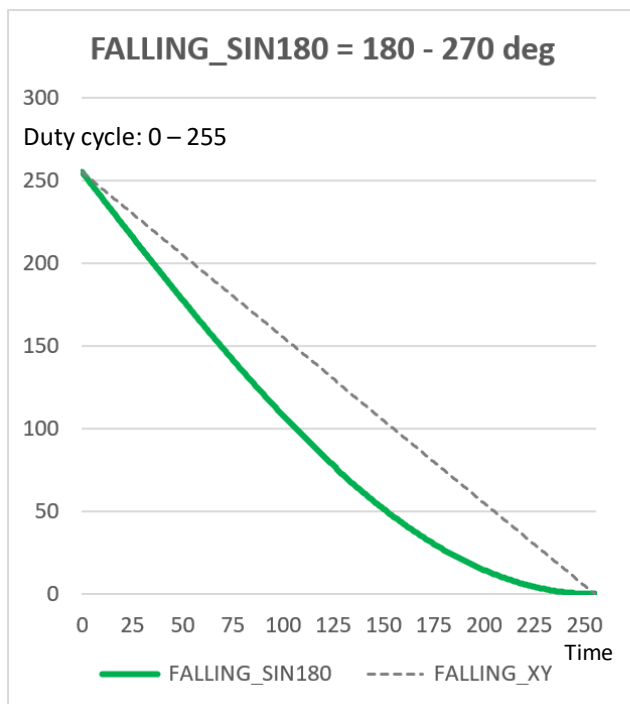
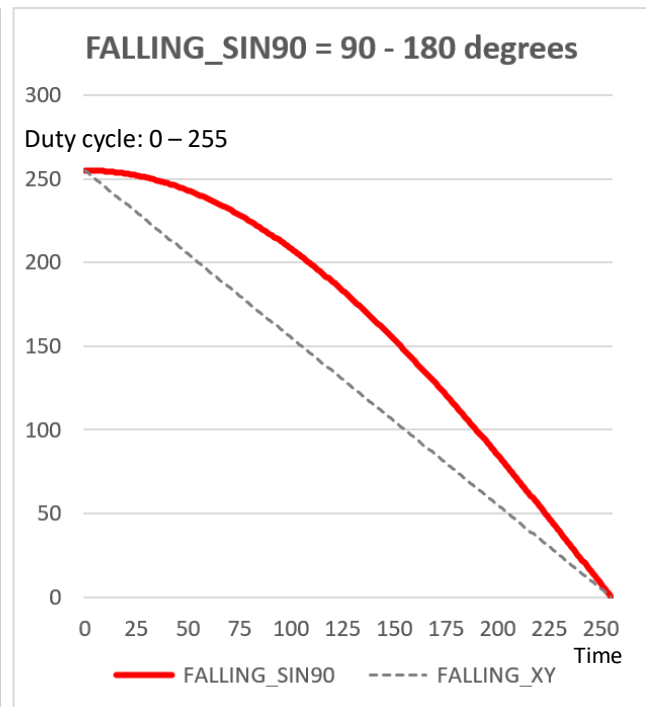
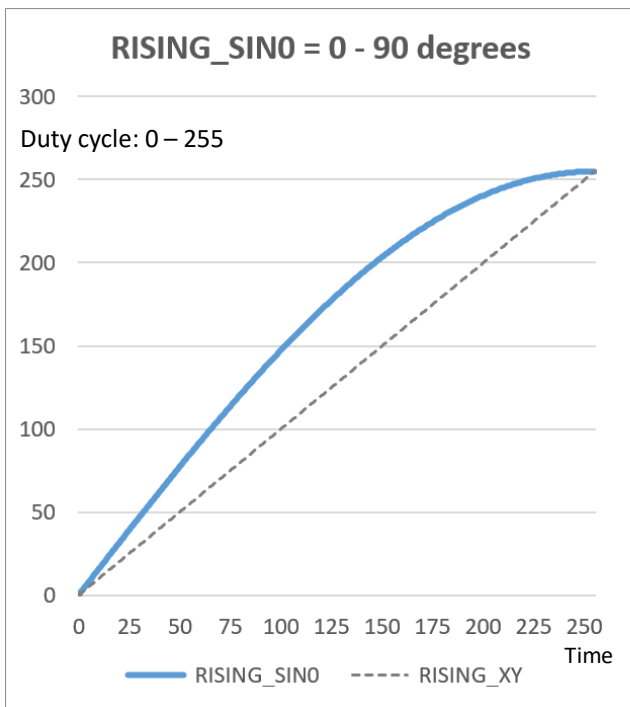


Arduino example:
DEMO video:

Sine curve

[Sine curve](#) is divided into four parts of each of 90 degrees, and distributed together in the range 0 – 360 degrees. In this way, it is possible to combine the individual parts into various new functions.





Arduino example:
DEMO video:

Pendulum operation using math PWM curves and H-bridges: <https://youtu.be/1i-cGc6Dk4E>

The different curves are named as global constants (MTD2A_const.h):

MIN_PWM_VALUE

MAX_PWM_VALUE

NO_CURVE

RISING_XY

RISING_LED

RISING_SM8

RISING_B05

FALLING_B05

RISING_BZ1

RISING_SIN0

FALLING_XY

FALLING_LED

RISING_SM5

RISING_B025

FALLING_B025

FALLING_BZ1

FALLING_SIN90

FALLING_SM8

RISING_E05

FALLING_E05

RISING_BZ2

RISING_SIN270

FALLING_SM5

RISING_E025

FALLING_E025

FALLING_BZ2

FALLING_SIN180

Please write if there is a desire for other PWM curves: MTD2A@jorgen-madsen.dk

Scaling

All curves can be scaled. If the value **beginValue** and the value **endValue** are defined, the entire curve is scaled within the time period **outputTimeMS**

Examples of scaling:

motor.activate (50, 200, RISING_XY);

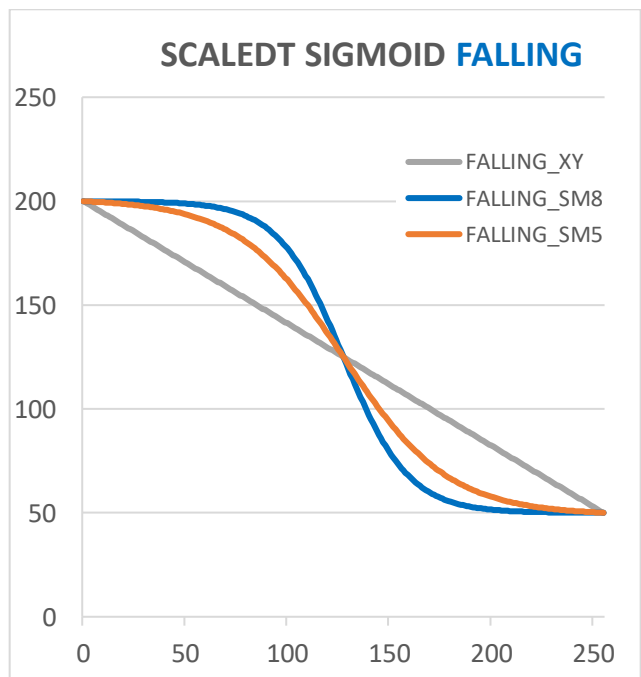
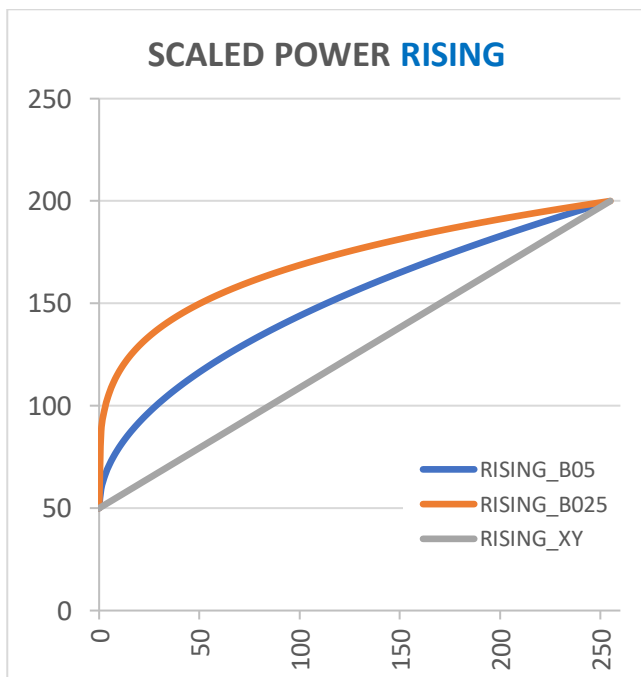
motor.activate (50, 200, RISING_B05);

motor.activate (50, 200, RISING_B025);

servo.activate (200, 50, FALLING_XY);

servo.activate (200, 50, FALLING_SM8);

servo.activate (200, 50, FALLING_SM5);



outputValue and PWM

Refer to the `outputValue` section for an introduction.

The individual PWM curves are calculated pointwise and evenly over `outputTimeMS` period, and with temporal steps determined by the `globalDelayTimeMS`. See the in-depth explanation of time management and synchronization in the document `MTD2A_base.pdf`.

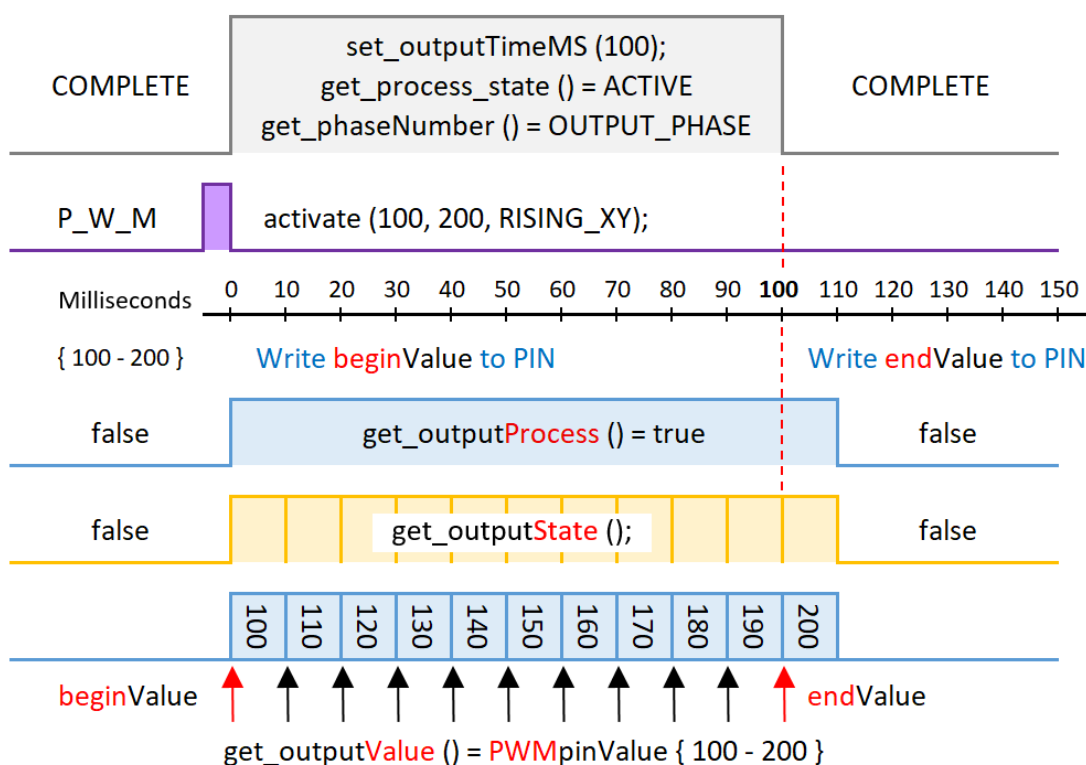
The individual calculated points on the curve can be read with the function: `object_name.get_outputValue ();`

`object_name.get_outputState ();` is `true` when there is **new data** to read. This means that if the same value occurs several times in a row, the function only returns `true` the first time the value occurs.

Whether or not to optimize is controlled with the function:

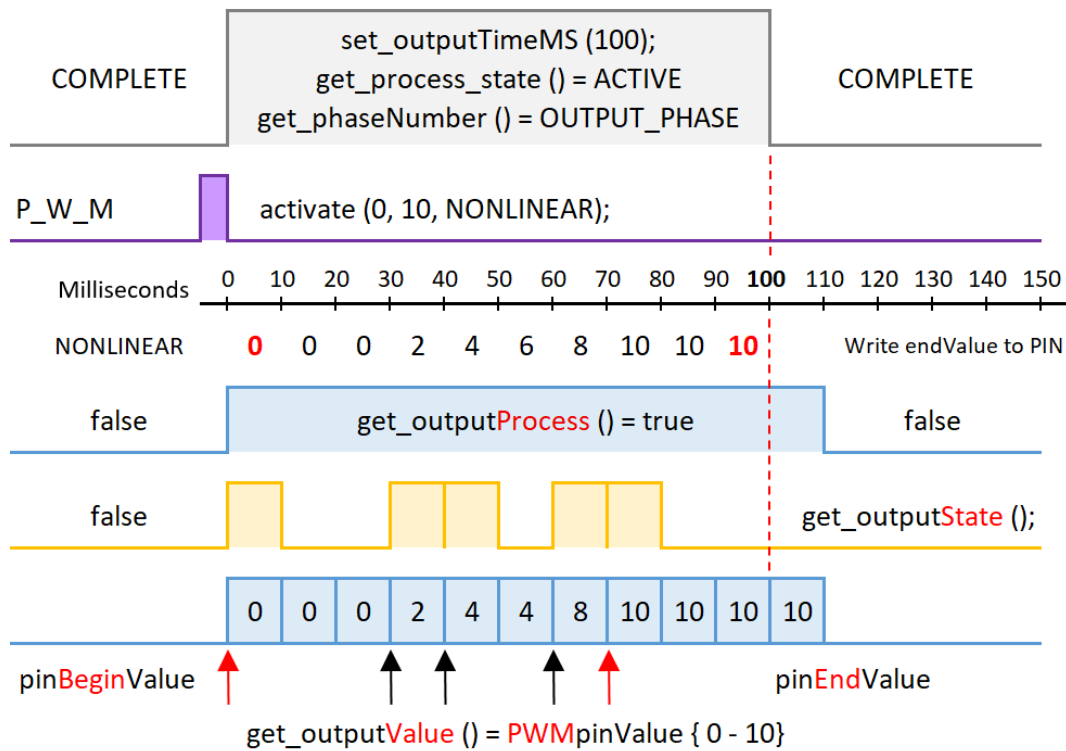
`object_name.set_outputOptimize ( {ENABLE | DISABLE} );`

The current configuration can be read with the function: `object_name.get_outputOptimize ();`



Example of linear PWM function `RISING_XY`

```
MTD2A_binary_output PWM_curve ("PWM_curve", 3000); // 3 seconds slow movement
Servo myServo;
// Initialize and loop etc.
if (PWM_curve.get_processState() == COMPLETE) {
    PWM_curve.activate (0, 180, RISING_XY); // begin 0 and end 180 degrees
}
if (PWM_curve.get_outputState () == true) {
    myServo.write( PWM_curve.get_outputValue() );
}
```



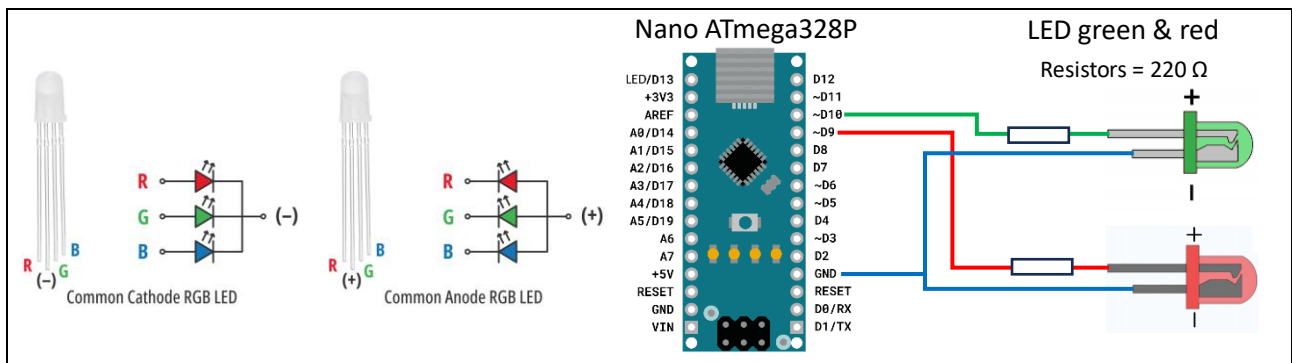
Example of *nonlinear* PMW Function *RISING_SM5*

```
MTD2A_binary_output PWM_curve ("PWM_curve", 3000); // 3 seconds slow movement
Servo myServo;
// Initialize and loop etc.
if (PWM_curve.get_processState() == COMPLETE) {
    PWM_curve.activate (0, 180, RISING_SM5); // begin 0 and end 180 degrees
}
if (PWM_curve.get_outputState () == true) {
    myServo.write( PWM_curve.get_outputValue() );
}
```

Examples of configuration

A multi-colored LED with a common **cathode** is activated by changing from LOW to HIGH. PWM 0 -> 255.

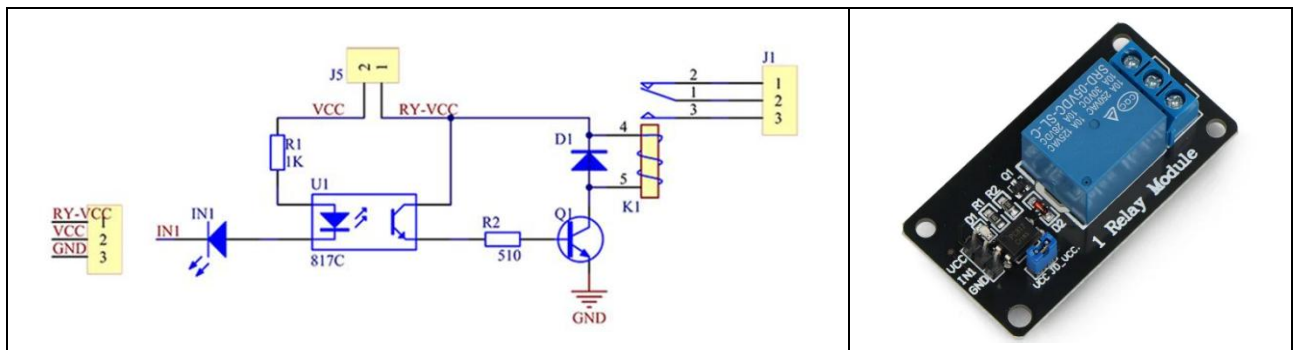
A multi-colored LED with a common **anode** is activated by changing from HIGH to LOW. PWM 255 -> 0.



Example of a green LED that waits for half a second and then lights up for half a second.

1. `MTD2A_binary_output green_LED ("Green LED", 500, 500);`
2. `green_LED.initialize (10);`
3. `green_LED.activate ();`
4. `MTD2A_loop_execute ();`

A standard relay with optocoupler that is activated by going from HIGH to LOW.



Example of an optocoupled relay that waits for half a second and is then activated for half a second.

All default values are the inverse of what is the default. Therefore, all parameters must be specified.

1. `MTD2A_binary_output opto_relay ("Opto relay", 500, 500, 0, BINARY, LOW, HIGH);`
2. `opto_relay.initialize (12, NORMAL, HIGH);`
3. `opto_relay.activate ();`
4. `MTD2A_loop_execute ();`

Alternatively, `INVERTED`, which inverts (reverses) all PIN output values.

1. `MTD2A_binary_output opto_relay ("Opto relay", 500, 500);`
2. `opto_relay.initialize (12, INVERTED);`
3. `opto_relay.activate ();`
4. `MTD2A_loop_execute ();`

Set get functions

Set functions	Green is default when no parameter	Comment
set_outputMode	({ BINARY P_W_M });	Select binary or PWM output
set_PWMcurveType	(PWM curve number/name)	Select PWM curve to use in PWM mode
set_outputOptimize	({ ENABLE DISABLE });	ENABLE : outputState is only when outputValue change from last outputValue DISABLE: outputState always true
set_pinWriteValue	({BINARY {HIGH LOW} / PWM {0-255} } , {BINARY P_W_M});	Write directly to output PIN (if enabled) Optional parameter. No default.
set_pinWriteToggle	({ ENABLE DISABLE });	Enable or disable PIN writing
set_pinWriteMode	({ NORMAL INVERTED });	Invert all output (PIN writing)
set_outputTimeMS	({ 0 - MAX_TIME_MS })	Set new output time period milliseconds
set_beginDelayMS	({ 0 - MAX_TIME_MS })	Set new begin delay time period millisec.
set_endDelayMS	({ 0 - MAX_TIME_MS })	Set new end delay time period millisec.
set_timers	(outputTimeMS, beginDelayMS, endDelayMS)	Set one, two or three timers milliseconds Optional parameters. No defaults.
set_outputTimer	({ STOP_TIMER RESET_TIMER })	Stop timer process immediately or reset
set_beginTimer	({ STOP_TIMER RESET_TIMER })	Stop timer process immediately or reset
set_endTimer	({ STOP_TIMER RESET_TIMER })	Stop timer process immediately or reset
set_loopActivate	({ ENABLE DISABLE })	Repeat the activation process until DISABLE
set_debugPrint	({ ENABLE DISABLE });	Enable print phase number and text
set_errorPrint	({ ENABLE DISABLE });	Enable error messages

Get functions	Comment
get_outputMode	(); return bool { BINARY P_W_M }
get_outputOptimize	(); return bool { ENABLE DISABLE }
get_PWMcurveType	(); Return uint8_t curve number
get_processState	(); return bool { ACTIVE COMPLETE }
get_pinWriteMode	(); return bool { NORMAL INVERTED }
get_pinWriteToggle	(); return bool { ENABLE DISABLE }
get_outputValue	() return uint8_t { HIGH LOW } / { 0 - 255 }
get_outputState	(); return bool { true false }
get_outputProcess	(); return bool { ACTIVE COMPLETE }
get_phaseChange	(); return bool { true false }
get_phaseNumber	(); return uint8_t { 0 - 4 }
get_setBeginMS	(); return uint32_t milliseconds
get_setOutputMS	(); return uint32_t milliseconds
get_setEndMS	(); return uint32_t milliseconds
get_loopActivate	({ ENABLE DISABLE })
get_reset_error	(); return uint8_t { 0-255 }

Operator overloading	Function
object_name & !objectName e.g. if (objectName) { activate };	bool phaseChange == true && phaseNumber == COMPLETE_PHASE
object_name_1 == object_name_2	bool processState_1 == processState_2
object_name_1 != object_name_2	bool processState_1 != processState_2
object_name_1 > object_name_2	bool processState_1 = ACTIVE & processState_2 = COMPLETE
object_name_1 < object_name_2	bool processState_1 = COMPLETE & processState_2 = ACTIVE

```
print_conf();
```

```
object_name.print_conf();
```

```
MTD2A_binary_output:
-----
objectName      : LED_1
processState    : ACTIVE
phaseText       : [3] End delay
debugPrint      : DISABLE
globalDebugPr   : DISABLE
errorPrint      : DISABLE
globalErrorPr   : ENABLE
errorNumber     : 0 OK
outputTimeMS    : 2000
beginDelayMS    : 0
endDelayMS      : 2000
outputMode      : PWM
PWMcurveType    : 0 (NO_CURVE)
PWMloopCount    : 200
beginValue      : 10
endValue        : 0
outputValue     : 255
pinNumber       : 9
pinWriteToggl   : ENABLE
pinWriteMode    : NORMAL
pinStartValue   : 0
PinWriteValue   : 0
setOutputMS     : 2014
setBeginMS      : 0
setEndMS        : 4015
```